

1.11 Test and Evaluate HCI (Evaluation Methods)

ٹیسٹنگ یہ دیکھتی ہے کہ پروڈکٹ ٹھیک چلتا ہے (بگز)؛ ایویلیوایشن یہ دیکھتی ہے کہ صارف کی ضرورت پوری ہوئی۔

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 14 of 15

1.11 Test and Evaluate HCI (Evaluation Methods)

LEARN IT

Testing checks if the product works properly (finding bugs). **Evaluation** determines whether it meets users' needs (usability). Together they improve the product before release.

A school portal can be "bug-free" (testing passed) and still unusable if students cannot find the date sheet (evaluation failed). You need both.

1.11.1 Evaluation methods

Method	Description
SUS (System Usability Scale)	A standard 10-item questionnaire used to calculate a score for how usable a system is.
NPS (Net Promoter Score)	Asks users: "How likely are you to recommend this app to a friend?" Measures user loyalty.
Formative evaluation	Done during development to collect early feedback and improve the design continuously.
Summative evaluation	Done at the end of development to measure overall success against the original goals.
Expert reviews	Usability specialists review the app against established design standards (heuristics).

Heuristics are agreed rules of good design (for example "show system status", "be consistent"). An expert review checks the product against those rules without waiting for a huge user survey.

WORKED EXAMPLE

Worked example: when to use each method on a school portal.

1. Week 2 of building: show the Lo-Fi to five students, change the menu. That is **formative** evaluation.
2. A teacher who has studied HCI checks the screens against standard rules (Is feedback visible? Are labels consistent?). That is an **expert review**.
3. After the portal is "finished", 30 students fill the 10 SUS questions. A low score means it is still hard to use, even if there are no crashes.
4. Ask: "On a scale of 0–10, would you recommend this portal to a friend in another college?" That is **NPS** (loyalty).
5. Compare the finished portal with the original goals (date sheet in one tap, bilingual fee notice). That is **summative** evaluation.

REMEMBER IT

Formative = forming it (during). Summative = summing it up (end). SUS = 10 questions on usability.
NPS = "Would you recommend it?" Expert = specialist vs heuristics.

COMMON MISTAKE

Mistake: Saying testing and evaluation are the same, or mixing formative with summative.

Fix: Testing = bugs. Evaluation = does it meet human needs. Formative is during; summative is at the end against original goals.

اردو میں سمجھیں

ٹیسٹنگ یہ دیکھتی ہے کہ پروڈکٹ ٹھیک چلتا ہے (بگز)؛ ایویلیوایشن یہ دیکھتی ہے کہ صارف کی ضرورت پوری ہوئی۔ SUS دس سوالوں والا استعمال کا اسکور ہے، NPS پوچھتا ہے کہ آپ یہ ایپ دوست کو کتنی تجویز کریں گے۔ فارمیٹو جانچ ترقی کے دوران ہوتی ہے، سمیٹو آخر میں اہداف کے خلاف۔ ماہر جائزہ ہیورسٹکس (Heuristics) پر ہوتا ہے۔

CHECK YOURSELF

1. What is the difference between testing and evaluation?
2. How many items are on the SUS questionnaire?
3. When is summative evaluation done?

Answers: 1. Testing finds bugs / checks the product works; evaluation checks it meets users' needs (usability). 2. 10. 3. At the end of development, against the original goals.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Testing asks:

- a. Does it work / any bugs?
- b. Is the logo pretty only
- c. What is Big O
- d. What is a FK

MCQ 2. Evaluation asks:

- a. Does it meet user needs / usability goals?
- b. Only CPU temp
- c. Only compile errors
- d. Only SQL syntax

MCQ 3. You need both because:

- a. A bug-free UI can still be unusable
- b. Bugs never matter
- c. Users never matter
- d. HCI is only art

MCQ 4. A usability goal example:

- a. First-time student pays fees in < 3 minutes
- b. More colours
- c. More menus
- d. More jargon

MCQ 5. Evaluation can be:

- a. With users or with experts/heuristics
- b. Only with the CEO's nephew
- c. Only after 5 years
- d. Only in assembly

MCQ 6. Do this:

- a. After a prototype, not only after full code
- b. Never
- c. Only on paper NAND
- d. Only on OSI diagrams

Extra short questions

1. Differentiate testing and evaluation in HCI.
2. Give a metric for each.
3. When is expert review useful?
4. Why include real students?

Extra long question

1. Plan an evaluation of the LMS assignment-upload: 2 test cases (bugs) and 2 evaluation questions (usability) with measures.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Testing: system behaviour/bugs. Evaluation: human fit/usability/need.
2. Test: crash rate. Eval: task success, time, SUS satisfaction, error types.
3. Early, cheap, catches obvious heuristic breaks before booking users.
4. They hold the real tasks and language.

Long

1. Test: 20MB PDF; no network. Eval: can a new student succeed unaided; time; satisfaction 1-5. n=8. What you would change if time > 5 min.